Non-Perturbative Floquet Engineering of the Toric-code Hamitonian

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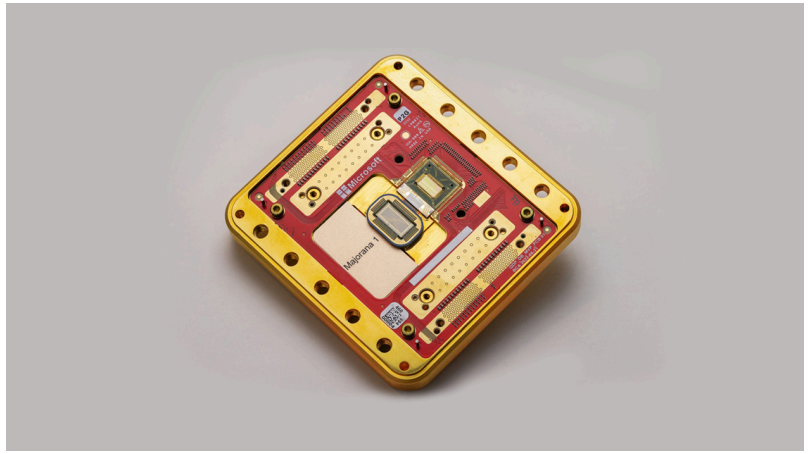
Main Contents

- Background(Quantum error correction and Toric code)
- Non-Perturbative Floquet Engineering of the Toric-code Hamitonian
- Prospects(PauliPropagation for larger system and introduction of self-correction)

Topological quantum computation

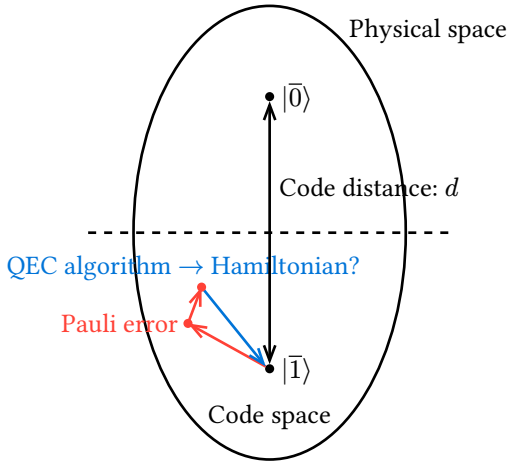
- Microsoft unveils Majorana 1, the world's first quantum processor powered by topological qubits



- Fault-tolerant prototype (FTP) based on topological qubits

QEC in neutral atoms array

- Logical qubits(k), physical qubits(n) and code distance(d)



- We require a large code distance(d), Logical qubits(k) and small physical qubits(n) to achieve fault-tolerant quantum computation.

Topological-protected Toric code(Kitaev, 2003)

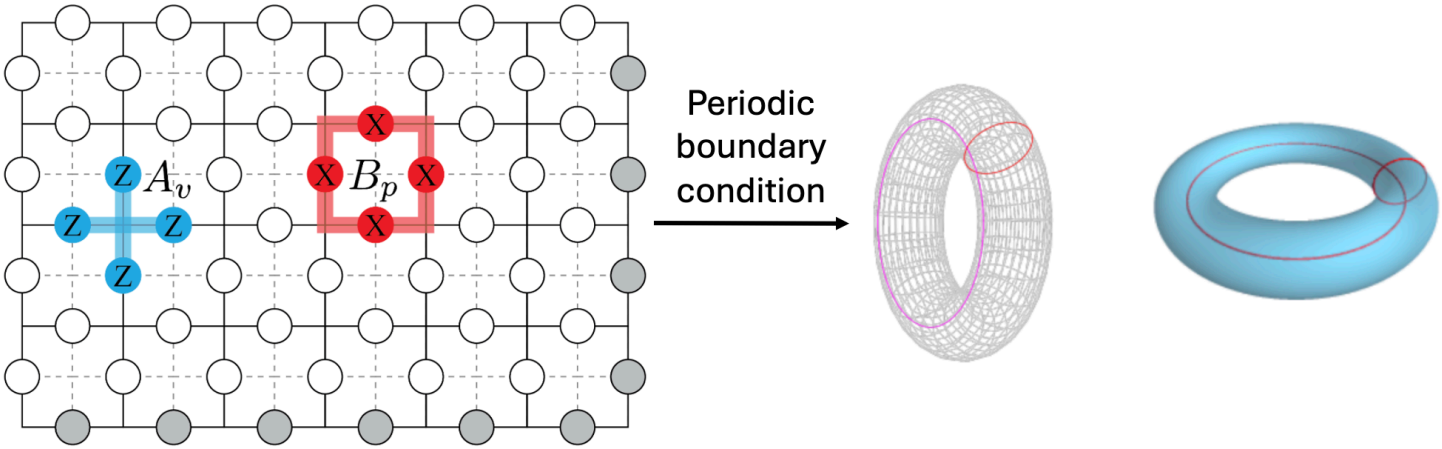


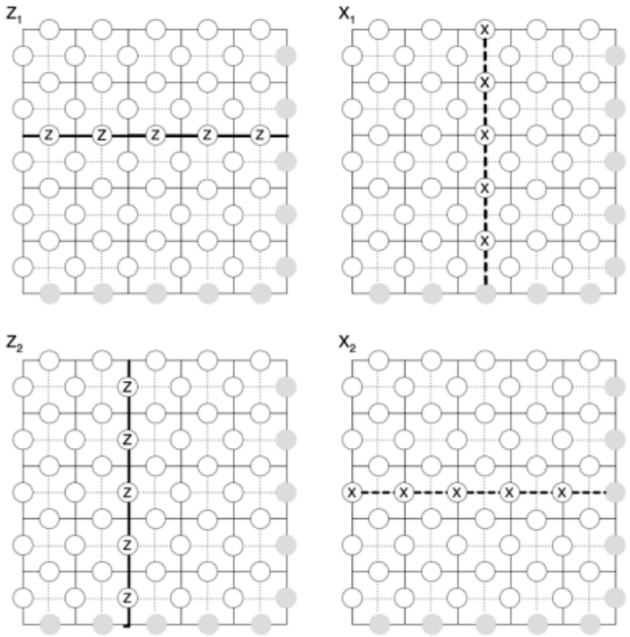
Figure 2: Left: Two loops cannot be deformed to a point or to each other. Right:square lattice with periodic boundary

$$H = -\sum_v A_v - \sum_p B_p, \left(A_v = \prod_{j\in \text{ star}(v)} Z_j, B_p = \prod_{j\in \text{ bdy}(v)} X_j \right)$$

$$[A_i, A_j] = [A_i, B_j] = [B_i, B_j] = 0$$

Topological-protected Toric code(Kitaev, 2003)

- Logical operators:



Construction of XZZX code(Petiziol et al., 2023)

arXiv:2211.09724

Non-Perturbative Floquet Engineering of the Toric-Code Hamiltonian and its Ground State

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We theoretically propose a quantum simulation scheme for the toric-code Hamiltonian, the paradigmatic model of a quantum spin liquid, based on time-periodic driving. We develop a hybrid continuous-digital strategy that exploits the commutativity of different terms in the target Hamiltonian. It allows one to realize the required four-body interactions in a nonperturbative way, attaining strong coupling and the suppression of undesired processes. In addition, we design an optimal protocol for preparing the topologically ordered ground states with high fidelity. A proof-of-principle implementation of a topological device and its use to simulate the topological phase transition are also discussed. The proposed scheme finds natural implementation in architectures of superconducting qubits with tuneable couplings.

Wen's model, the XZZX surface code(Acharya et al., 2022)

$$H_\omega = -J \sum_{i,j} \hat{P}_{i,j} \left(\hat{P}_{i,j} = \hat{X}_{i,j} \hat{Z}_{i,j+1} \hat{Z}_{i+1,j} \hat{X}_{i+1,j+1} \right)$$

- Standard surface code + Hardmard the qubit shown in green circle = the XZZX code.

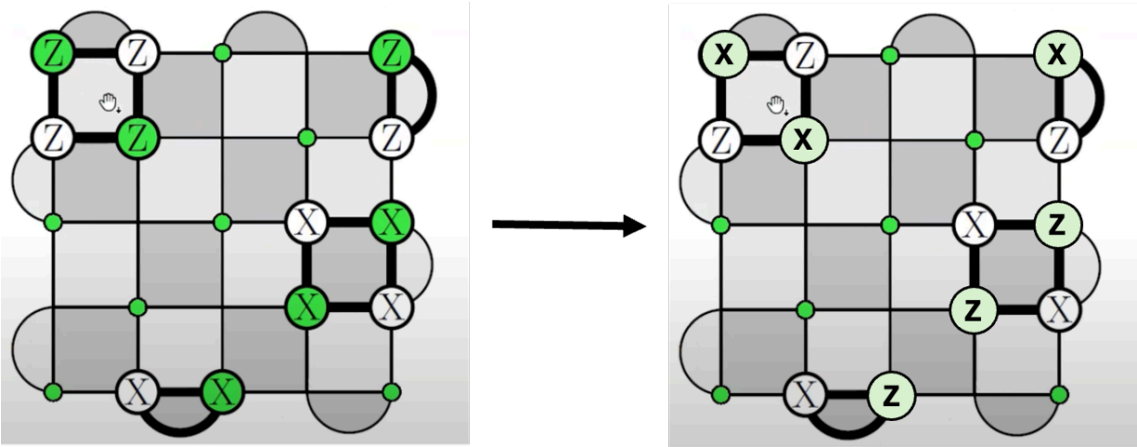

Figure 5: XZZX code

Surface code(Bombin & Martin-Delgado, 2007; Kitaev, 2003)

- Formation of rotated surface code:(Kang et al., n.d.) [surfacecode]

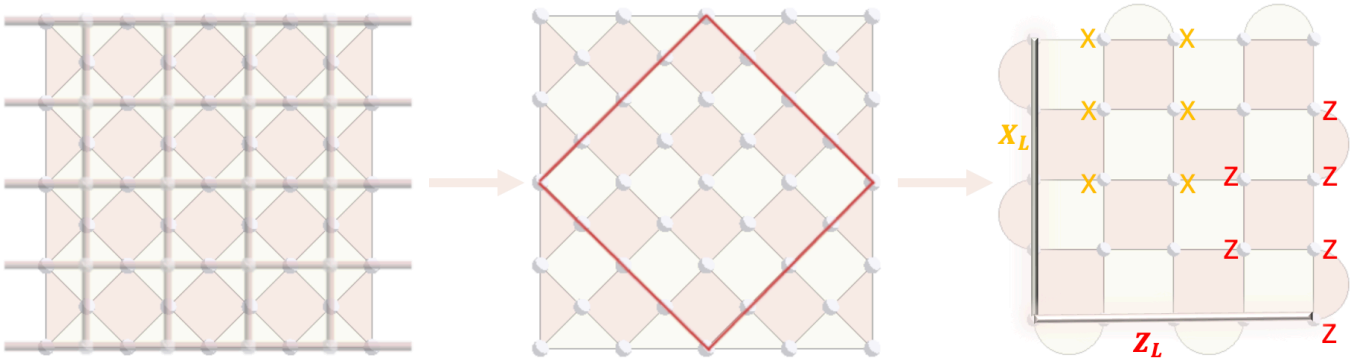

Figure 6: Surface code.

- logical operators: vertical Z chain and horizontal X chain.

XZZX code

- Single-qubit Pauli noise channel(Biased)

$$\varepsilon(\rho) = (1-p)\rho + p_x X\rho X + p_y Y\rho Y + p_z Z\rho Z$$

- Dominant noise type: dephasing (Z errors)(Grimm et al., n.d.; Pla et al., n.d.)

– XZZX code: “every single stabilizer in the new code can detect odd parity of Z errors nearby”.

– Original code: “Only half the X stabilizers could detect Z errors”.

Wen's model, the XZZX surface code(Acharya et al., 2022)

- threshold for the XZZX code(left) and the CSS code(right)(Tuckett et al., 2020)

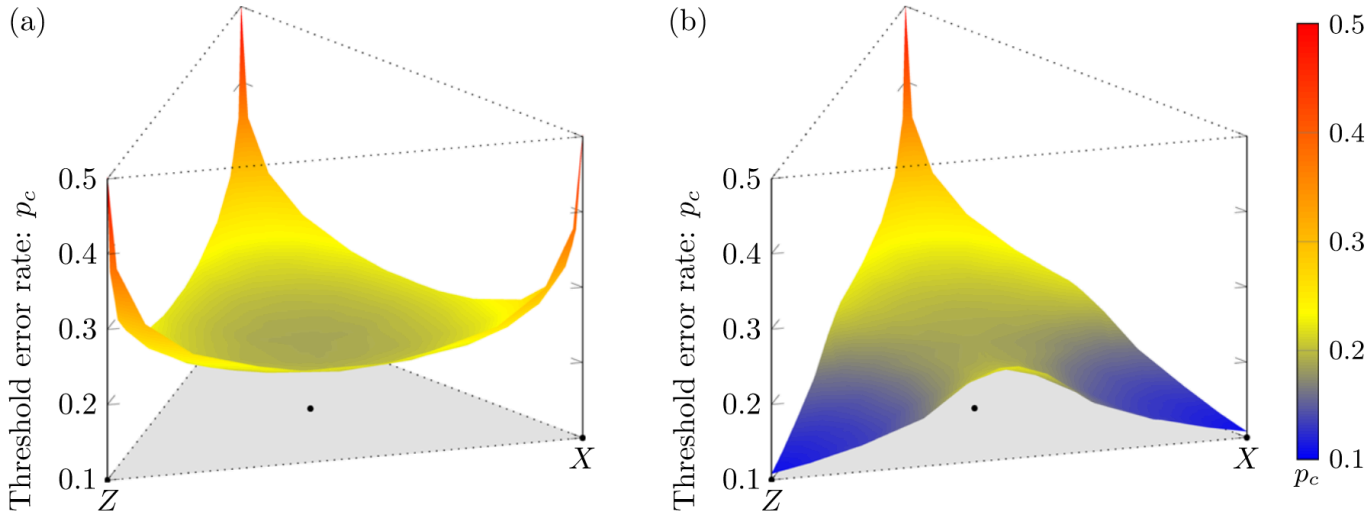

Figure 7: Optimal code-capacity thresholds over all single-qubit Pauli channels

Non-Perturbative Floquet Engineering of the Toric-code Hamitonian

Motivation

- Realizing the Toric-code Hamiltonian in 2D lattice.
  - Clean four-spin interactions.
- Methods:
  - Floquet engineering.
  - Trotterization.
- It is promising in the self-correcting machanism smaller than 4D quantum system.

Model

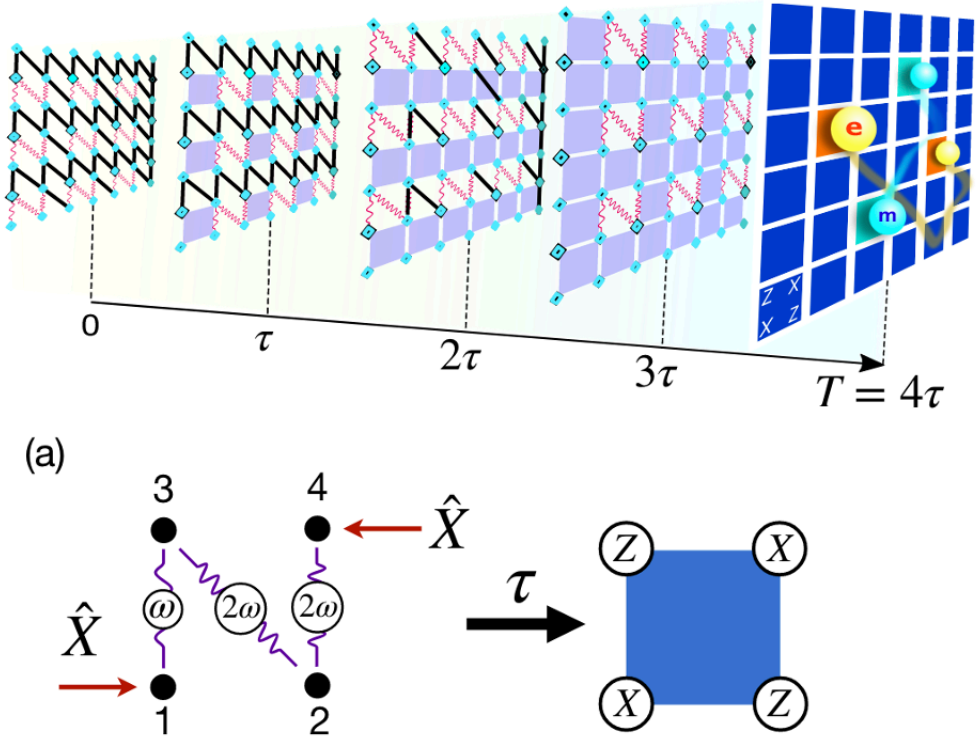

Figure 8: Realizing the Toric-code Hamiltonian in 2D lattice through Floquet engineering and trotterization.

Model

Hamitonians of driven/target system

- Driven system: $\hat{H}(t) = \hat{H}_1(t) + \hat{H}_2(t)$,

$$\hat{H}_1(t) = -g_{\alpha\beta} \sum_{\langle\alpha,\beta\rangle} \left(a_\alpha^\dagger a_\beta + a_\beta^\dagger a_\alpha\right) \to \sum_{\langle\alpha,\beta\rangle} g_{\alpha\beta}\left(\hat{X}_\alpha \hat{X}_\beta + \hat{Y}_\alpha \hat{Y}_\beta\right)$$

- Target system: Wen's plaquette model(Brown et al., 2011)(=Kitaev's model)(Nussinov & Ortiz, 2009) on a square lattice

$$\hat{H}_\omega = -J \sum_{i,j} \hat{P}_{i,j}$$

$$\hat{P}_{i,j} = \hat{X}_{i,j} \hat{Z}_{i,j+1} \hat{Z}_{i+1,j} \hat{X}_{i+1,j+1}$$

Why simulating Wen's Model

- Wen's formulation:

  – a promising surface code dessign against biased noise(Acharya et al., 2022; Bonilla Ataides et al., n.d.).

- Method: Finding control functions $g_{\alpha\beta}$.

Floquet engineering four-spin interactions

Central difficulty

- clean four-spin interactions.

Floquet engineering

- Considering a foundamental four-spin plaquette.
- For a time step τ, effective Hamiltonian $H_{i,j}$ at τ is derived through Magnus expansion.

$$\exp\left(-i\hat{H}_{i,j}\tau\right) = \mathrm{T}\exp\left(-i\int_0^\tau \mathrm{d}t\hat{H}(t)\right)$$

- Search for periodic $g_{\alpha\beta}(t) \to H_{i,j} \approx -JP_{i,j}$.

Single plaquette model

- Magnus expansion(Bukov et al., 2015):

$$\hat{H}_{\mathrm{i,j}} = \frac{1}{\tau}\sum_{k=1}^{\infty}\hat{\Omega}_k(t)$$

$$\hat{\Omega}_1(t) = \int_0^{\tau}\hat{H}(t)\mathrm{dt} \mid \mid \hat{\Omega}_2(t) = -\frac{1}{2}\int_0^{\tau}\int_0^{t}\left[\hat{H}(t),\hat{H}(t')\right]\mathrm{dt'dt}$$

$$\hat{\Omega}_3(t) = \frac{1}{6}\int_0^{\tau}\mathrm{dt}_1\int_0^{t_1}\mathrm{dt}_2\int_0^{t_3}\mathrm{dt}_3\left[\hat{H}(t_1),\left[\hat{H}(t_2),\hat{H}(t_3)\right]\right]$$

$$\hat{X}_1\hat{Z}_2\hat{Z}_3\hat{X}_4 = [\hat{X}_1\hat{X}_2,\left[\hat{X}_2\hat{X}_3+\hat{Y}_2\hat{Y}_3,\hat{X}_3\hat{X}_4\right]$$

Single plaquette model

Driving functions

$$g_{13} = g_{13}\cos(\omega t)$$

$$g_{23} = g_{13}\cos(2\omega t)$$

$$g_{24} = g_{13}\cos(2\omega t)$$

$$\omega = 2\pi/\tau$$

- It is time symmetric : $g_{\mathrm{kk'}}(t) = g_{\mathrm{kk'}}(\tau - t)$

Single plaquette model

Single plaquette model

- Strong resonant X-drive: $\hat{H}_2 = \Omega_1 \hat{X}_1 + \Omega_4 \hat{X}_4, (\Omega_{1,4} \gg g_{\mathrm{kk'}(t)})$.

(Penalizing the process induced by $\hat{Y}\hat{Y}$ coupling.)

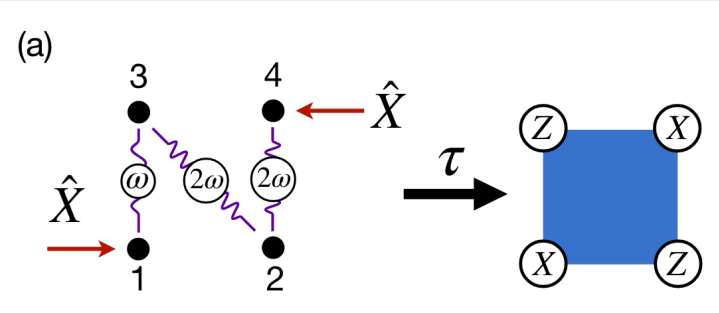

Figure 9: Sketch of the driving scheme for realizing a fourspin term in a single four-spin plaquette.

- $\hat{H}(t) = \Omega_1 \hat{X}_1 + \Omega_4 \hat{X}_4 + \sum_{\langle k,k' \rangle} g_{\mathrm{kk'}} \left(\hat{X}_k \hat{X}_{k'} + \hat{Y}_k \hat{Y}_{k'} \right)$

Single plaquette model

- Eliminating YY term:

$$\Omega_1\hat{X}_1 + \Omega_4\hat{X}_4 + g_{13}(t)\left(\hat{X}_1\hat{X}_3 + \hat{Y}_1\hat{Y}_3\right) + g_{23}(t)\left(\hat{X}_2\hat{X}_3 + \hat{Y}_2\hat{Y}_3\right) + g_{24}(t)\left(\hat{X}_2\hat{X}_4 + \hat{Y}_2\hat{Y}_4\right)$$

$$\Omega_1 = \Omega_4 = M * \omega$$

- Rotating frame with respect to X_1, X_4 terms:

$$g_{13}(t)\left(\hat{X}_1\hat{X}_3 + e^{iM\omega t\hat{X}_1}\hat{Y}_1 e^{-iM\omega t\hat{X}_1}\hat{Y}_3\right) + g_{23}(t)\left(\hat{X}_2\hat{X}_3 + \hat{Y}_2\hat{Y}_3\right)$$

$$+g_{24}(t)\left(\hat{X}_2\hat{X}_4 + \hat{Y}_2 e^{iM\omega t\hat{X}_4}\hat{Y}_4 e^{-iM\omega t\hat{X}_4}\right)$$

Single plaquette model

Effective Hamitonian

- All possible operators ← dynamical Lie algebra of the control system(Appendix B3) (D'Alessandro, 2007).
- Effective Hamiltonian

$$\hat{H}_{i,j} = \sum_{O \in L} c_l \hat{O}_l$$

- Challenge: Single out the desired term.

single plaquette model

Strong and clean 4-body-interaction

- Method: Optimize $g_{kk'}, \Omega_j$ via exact numerical simulation in single-plaquette model rather than high-frequency expansion.
- The four-body interaction is the dominant term:

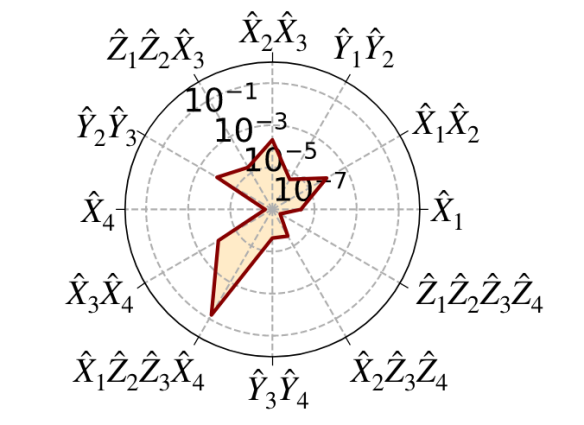

Figure 10: Structure of the lattice.

single plaquette model

Effective Hamiltonian

- Effective Hamiltonian

$$\hat{H}_{i,j} = -J\hat{P}_{i,j} + \varepsilon\hat{V}_{i,j}$$

The extra terms are contained in $\hat{V}_{\mathrm{i,j}}$ and $\varepsilon/J = 1.5 \cdot 10^{-3}$.

Full lattice sequence

first-order Floquet-Trotter sequence

- Devide the Hamitonian as the sum of four parts:

$$\hat{H}_k = \sum_{(i,j)\in G_k} \hat{H}_{i,j}$$

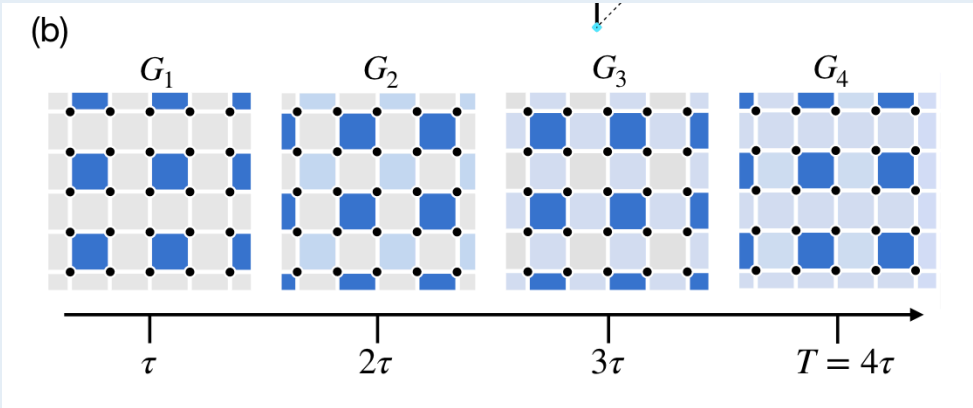

Figure 11: Trotter sequence to realize the Hamiltonian on the whole lattice.

Full lattice sequence

- The propogator:

$$\hat{U}(T) = \prod_{k=1}^{4} \prod_{(i,j)\in G_k} \hat{U}_{\mathrm{ij}(\tau)}$$

where $U_{\mathrm{ij}}(\tau) = e^{-i\tau \hat{H}_{i,j}}$ The product of propogators $\hat{U}_k = e^{-i\hat{H}_k \tau}$ belonging to two different groups do not commute and can be estimated by expontial product formula.

$$\prod_k \hat{U}_k = \exp\left(-i\tau \sum_k \hat{H}_k + \frac{\tau^2}{2} \sum_{j<k} \left[\hat{H}_j, \hat{H}_k\right] + \ldots\right)$$

- In effective Hamitonian, the term $P_{i,j}$ commutes with each other, so the equation above is used to calculate error terms. $\rightarrow \hat{H}_{\mathrm{eff}} = \hat{H}_\omega + \widehat{\mathrm{Err}}$

Simulation in 5x5 lattice

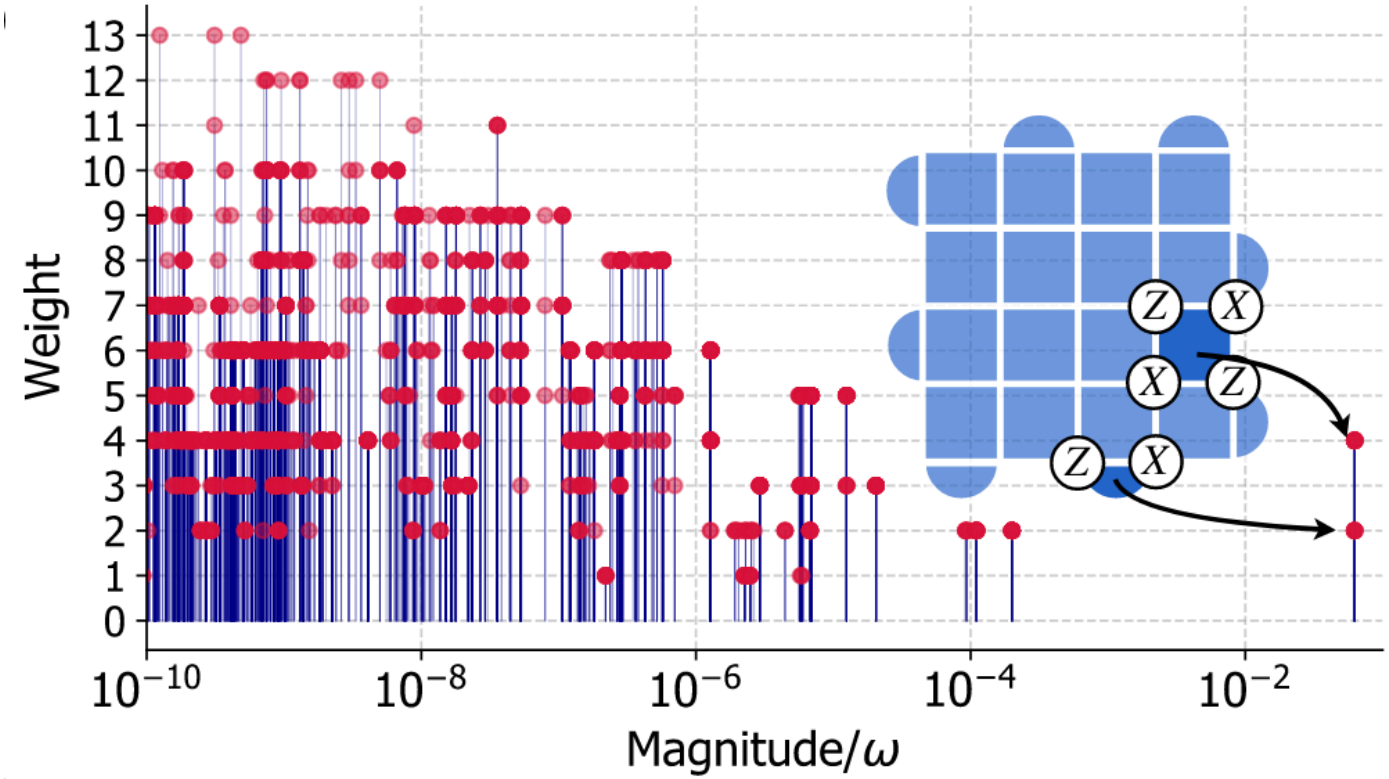

Figure 12: 5x5 lattice with mixed boundary conditions.

Prospects

- Main target: Proving the scaling performance of the Floquet engineering XZZX code.
- Method: Simulation of lager system with PauliPropogation.jl package.
- package: PauliPropogation.jl(https://github.com/MSRudolph/PauliPropagation.jl/tree/main)

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